

Course Title: Hydrometallurgy

MET 05103

Course Description:

This course covers the principles, methods, and applications of hydrometallurgical processes used in the extraction of metals from ores, concentrates, and recycled or waste materials using aqueous chemistry.

Course Outline

Module 1: Introduction to Hydrometallurgy

- Overview of extractive metallurgy
- Comparison: Hydrometallurgy vs. Pyrometallurgy vs. Electrometallurgy
- Historical background and industrial significance

Module 3: Leaching

- Principles and mechanisms
- Types of leaching: heap, tank, pressure, in-situ
- Leaching agents: acids, alkalis, cyanide, ammonia, thiourea, thiosulfate
- Factors affecting leaching efficiency

Module 4: Solution Purification and Concentration

- Solid-liquid separation techniques
- Precipitation methods
- Solvent extraction (SX)
- Ion exchange
- Cementation

Module 5: Metal Recovery Techniques

- Electrowinning
- Electrorefining
- Precipitation of metals
- Crystallization

Module 7: Environmental and Economic Aspects

- Waste treatment and recycling
 - Effluent control
 - Economic analysis and process selection
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